

The Photonic Holonet

A Single Self-Entangled Photon as Universal Computer, Universal Network, and Clock
on the $W(3, 3)$ Substrate

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Abstract

We present a complete, machine-verified architecture in which one self-entangled photon, traversing one finite geometry, is simultaneously the hardware, the software, and the network of a universal quantum computer. The geometry is the symplectic generalized quadrangle $W(3, 3)$: forty points, forty lines, automorphism group $\mathrm{Sp}(4, \mathbb{F}_3)$ of order 51840. The photon carries $W(3, 3)$ twice over — once as the forty Witting rays of its path–polarization space $\mathbb{C}^4$ (states), once as the forty Pauli displacement classes of its past–future time-bin space $\mathbb{C}^9$ (operators) — and the two carriers realize the same incidence geometry, making the machine’s states and gates one set of objects. Every architectural layer is a theorem with an executable witness: the preparation circuit (two Clifford gates of tabletop optics), the routing fabric (an atlas of 540 hypercube charts glued along the 1620 apartments of the Tits building), the protected memory (the Steinberg module of dimension 81), the mirror middleware (a 2160-slot D_{12} transport bus whose full Clifford lift factors as $24 \cdot 45 \cdot 48 = 51840$), the fuel (the matter shell is exactly the magic sector, with exact Kochen–Specker budget $36/40$ and contextual fraction $1/10$), the clock (an internal $\mathbb{Z}_{12}$ gauge clock, two external references $\mathbb{Z}_7, \mathbb{Z}_{13}$, and an irrational Boerdijk–Coxeter drive realizing a discrete time quasicrystal), and universality (the photon’s own tritter, phase plate, and modulator generate the complete Clifford group; the matter shell supplies the magic). No fitted parameters appear anywhere: every constant is substrate arithmetic at $q = 3$.

Contents

1	Overture: the machine is the geometry is the carrier	2
2	The substrate	3
2.1	The two carriers and the operator–operand duality	3
2.2	The five vacua and the Schläfli geography	3
3	The carrier: how to self-entangle a photon	3
3.1	Stage A: spatial (one element)	4
3.2	Stage B: temporal (two Clifford gates)	4
4	The network: atlas, building, and routing	4
5	The middleware: tomatope mirrors and the runtime bus	5

6	The software: braids, teleported gates, universality	6
6.1	Exact braid words	6
6.2	The photon is its own gate	6
6.3	The Universality Theorem	6
7	The fuel: matter equals magic	7
8	Memory, protection, and the immune system	7
9	Time: three clocks and a quasicrystal	7
10	Synchronization: beacons and schedules	8
11	Build sheet and falsifiable witnesses	8
12	Implications, corrections, and the ethos	8
A	Verification ledger	9

1 Overture: the machine is the geometry is the carrier

Classical architecture separates three things: the processor that acts, the program that directs, and the network that connects. The thesis of this paper is that at the level of a single photon on a finite symplectic geometry the three are one object, and that this is not a metaphor but a chain of theorems.

Principle 1.1 (Identity of the three layers). *Transporting the photon through the mesh is applying a gate is routing a packet. One physical process; three descriptions.*

The identification rests on three exact pillars, each proved by explicit computation in the project repository (every theorem below carries its witness script):

1. **Operator–operand duality** (§2.1): the same $W(3,3)$ is drawn by the photon’s *states* (orthogonality of the 40 Witting rays in $\mathbb{C}^4$) and by its *operators* (commutation of the 40 two-qutrit Pauli classes on $\mathbb{C}^9$).
2. **Routing–gate identity** (§4): the mesh is an atlas of 540 three-dimensional hypercube charts whose native XOR routing moves are exactly the register operations, glued along the 1620 apartments of the Tits building of $\mathrm{Sp}(4, \mathbb{F}_3)$.
3. **Mirror middleware** (§5): the cyclic selector clock and the mirror transport bus are different 2160 worlds. The photon is clocked by the C_{12} side but routed by the D_{12} side, whose full Clifford lift is the exact product $24 \cdot 45 \cdot 48$.
4. **Universality from optics** (§6.3): the photon’s own beam-splitting elements generate the complete Clifford group (symplectic closure = 51840, exact), and its matter shell is precisely its magic supply.

2 The substrate

Definition 2.1 ($W(3, 3)$). Let $V = \mathbb{F}_3^4$ with the alternating form $\langle u, v \rangle = u_1v_3 - u_3v_1 + u_2v_4 - u_4v_2$. The points of $W(3, 3)$ are the 40 projective points of V ; the lines are the 40 totally isotropic projective lines. Each point lies on 4 lines and each line carries 4 points; collinearity defines the strongly regular graph $\text{SRG}(40, 12, 2, 4)$.

The full automorphism group is $\text{Sp}(4, \mathbb{F}_3)$, of order $51840 = |W(E_6)|$, acting with permutation rank 3. The substrate primitives are

$$q = 3, \quad \lambda = 2, \quad \mu = 4, \quad k = 12, \quad f = 24, \quad g = 15, \quad \Phi_6 = 7, \quad \Phi_4 = 10, \quad \Phi_3 = 13,$$

and every numerical claim in this paper reduces to arithmetic in these.

2.1 The two carriers and the operator–operand duality

Theorem 2.2 (Two realizations, one geometry). (i) The 40 Witting rays in $\mathbb{C}^4$ (the photon’s path $\otimes$ polarization space) have orthogonality graph isomorphic to the collinearity graph of $W(3, 3)$. (ii) The 40 projective classes of two-qutrit Pauli displacements $D(v) = X^a Z^b \otimes X^c Z^d$ on $\mathbb{C}^9$ (the photon’s past $\otimes$ future time-bin space) have commutation graph equal to the same $W(3, 3)$, via $D(u)D(v) = \omega^{(u,v)} D(v)D(u)$. Witnesses: `bt817_self_entangled_photon_atlas.py`, `bt821_operator_operand_duality.py`.

Thus a point of the substrate is simultaneously a pure state the photon can occupy and a unitary the photon can enact; a line is simultaneously a measurement context (orthonormal tetrad) and a stabilizer group (maximal commuting Pauli set with its 9-state joint eigenbasis). The $360 = 40 \times 9 = f \cdot g$ two-qutrit stabilizer states are the joint eigenbases of the forty lines.

2.2 The five vacua and the Schläfli geography

Every maximal subgroup class of $\text{PSp}(4, 3)$ decomposes the substrate in its own canonical way; the maximal indices are the classical inventory of the 27 lines on a cubic surface.

index	subgroup	points	lines	vacuum	cubic surface
27	$2^4:A_5$	[40]	[40]	homogeneous (icosahedral)	27 lines
36	S_6	[40]	[10, 30]	spread fibration	36 double-sixes
40	parabolic	[1, 12, 27]	[4, 36]	holonet point vacuum	trihedra I
40	parabolic	[4, 36]	[1, 12, 27]	dual line vacuum	trihedra II
45	$(2T \times 2T):2$	[8, 32]	[16, 24]	binary polar	45 tritangents

The physical decomposition $40 = 1 + 12 + 27$ (self + gauge shell + matter shell) is the *choice of the point-parabolic vacuum* among five inequivalent readings; the icosahedral vacuum sees a featureless substrate. The platonic ladder threads the table: the binary tetrahedral group $2T$ (the 24-cell) lives in every polar-pair stabilizer, the real octahedral group $O_h (= S_4 \times \mathbb{Z}_2, \text{ order } 48)$ is the chart symmetry of §4, and the binary icosahedral group $2I = \text{SL}(2, 5)$ (the 600-cell) is the core of every spread stabilizer, with point orbits [20, 20] and line orbits [10, 30] echoing the Boerdijk–Coxeter decomposition $600 = 20 \times 30$. *Witnesses:* `bt808-bt813`.

3 The carrier: how to self-entangle a photon

Entanglement is a relation between tensor factors; nothing requires the factors to be distinct particles. Self-entanglement is entanglement between one photon’s own registers.

3.1 Stage A: spatial (one element)

A diagonal photon meets a polarizing beam splitter: $\frac{1}{\sqrt{2}}(|H\rangle + |V\rangle) \mapsto \frac{1}{\sqrt{2}}(|H, a\rangle + |V, b\rangle)$. Path and polarization are now maximally entangled *within* the photon; this two-qubit $\mathbb{C}^4$ is the Witting carrier of Theorem 2.2(i).

3.2 Stage B: temporal (two Clifford gates)

A tritter (symmetric 3-port coupler) *is* the qutrit Fourier gate F_3 ; a three-bin delay ladder $(0, \tau, 2\tau)$ defines past and future registers; one electro-optic modulator applies $CX_{p \rightarrow f}$ conditioned on bin index. The temporal Bell qutrit

$$|\Omega\rangle = CX_{p \rightarrow f}(F_3 \otimes I)|0\rangle_p|0\rangle_f = \frac{1}{\sqrt{3}} \sum_{j=0}^2 |j\rangle_p |j\rangle_f$$

is verified exactly; the photon is entangled with its own future. Inserting any U in the future arm, the recombination visibility is the *trace-Choi witness*

$$V(U) = \frac{|\text{Tr } U|}{3}, \quad V(F_3) = \frac{1}{3}, \quad V(X) = V(Z) = 0,$$

so the photon implements the Choi–Jamiołkowski isomorphism on itself: it measures channels against its own past. *Witness: `bt820_self_entanglement_protocol.py`.*

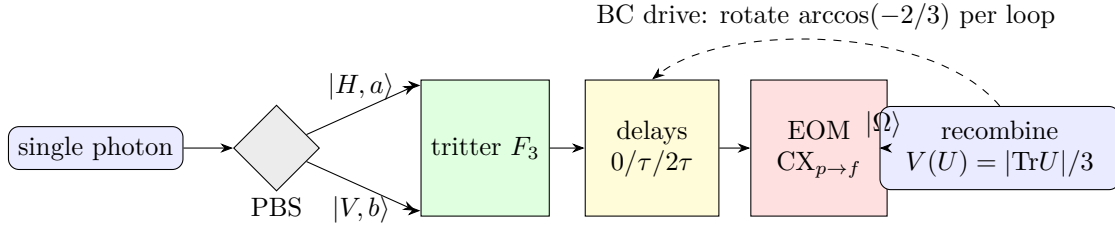


Figure 1: Preparation schematic. One photon, four registers. The PBS entangles path with polarization (Stage A); the tritter–delay–EOM chain prepares the temporal Bell qutrit $|\Omega\rangle$ (Stage B); the dashed loop is the irrational Boerdijk–Coxeter drive of §9.

4 The network: atlas, building, and routing

Theorem 4.1 (The hypercube atlas). *$W(3, 3)$ contains exactly 540 skew line pairs. For each pair the non-collinearity graph on its 8 points is the cube graph Q_3 , with full symmetry group O_h of order 48 (these are the only order-48 stabilizers; the binary $2O$ does not occur). Each chart admits an exact $\mathbb{F}_2^3$ Gray-code addressing in which chart edges are unit XOR moves and the antipode of a vertex is its unique collinear partner. Every $W(3, 3)$ nonedge is a chart edge in exactly $12 = k$ charts; every edge is an antipode pair in exactly $9 = q^2$. Witnesses: `bt773`, `bt777`, `bt778`.*

Theorem 4.2 (The fabric is the building). *The chart-adjacency web (540 nodes, two charts adjacent when one’s axis is a disjoint transversal pair of the other) is 6-regular with exactly 1620 edges, and these edges are in canonical bijection with the 1620 apartments of the Tits building of $\text{Sp}(4, \mathbb{F}_3)$. The web has diameter 5; its adjacency module contains the Steinberg representation twice and presses against the Ramanujan bound only on a 15-dimensional sentinel eigenspace. Witnesses: `bt777`, `bt779`, `bt744`.*

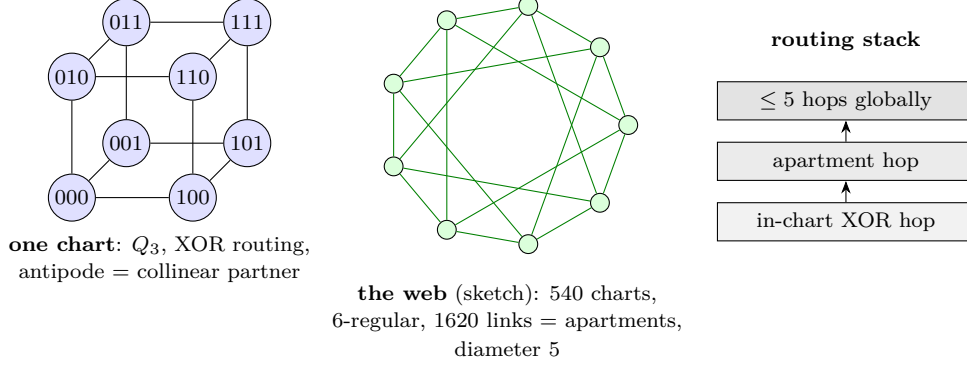


Figure 2: Network schematics. Left: a single chart with its $\mathbb{F}_2^3$ addressing (XOR routing native). Center: the 540-chart web whose links are the apartments of the Tits building (sketched as a circulant fragment). Right: the two-level routing stack.

Routing a packet means choosing XOR moves inside charts and apartment hops between them; since chart moves *are* register operations (§6), routing is computation. At scale the recursion is the holonet ladder: 40^n cores at level n , the same $\text{SRG}(40, 12, 2, 4)$ at every level, with the level-0 chart's $1 + 12 + 27$ degree split realized as self pole, gauge shell, and matter shell of each node.

5 The middleware: toмотope mirrors and the runtime bus

The atlas contains two different 2160-element boundary worlds. They must not be collapsed. The rectangle side is cyclic: its stabilizer is C_{12} and it supplies the selector's phase clock. The chart-transversal side is dihedral: its stabilizer is D_{12} and it supplies the mirror transport bus. Equal cardinality is the red herring; different stabilizer structure is the theorem.

Theorem 5.1 (The D_{12} mirror bus). *The 540 cube charts carry exactly four common-transversal slots each, so the global chart-transversal space has $540 \cdot 4 = 2160$ slots. The map*

$$(\text{chart}, \text{common transversal line}) \mapsto (\text{chart}, \text{antipode pair cut out by that transversal})$$

is an equivariant bijection onto the atlas antipode-slot space. Its projective stabilizer has order 12 and GAP identifies it as D_{12} , not C_{12} . Witness: `bt815_global_2160_transversal_gset.py`.

Locally, a skew-line chart has four common isotropic transversals. Read those four transversals as the four vertex carriers of the toмотope middle layer. Each transversal is a K_4 , and each K_4 has three opposite-edge axes and four triangular faces. Thus the chart carries

$$4 \text{ vertices}, \quad 4 \cdot 3 = 12 \text{ edge labels}, \quad 4 \cdot 4 = 16 \text{ face labels},$$

and the middle incidence packet has

$$4 \cdot 3 \cdot 4 = 48$$

edge-face blocks, with the doubled base/shadow fiber restoring 192 flags. This is the local $\langle r_0, r_3 \rangle$ toмотope block layer, not an analogy to a toroidal polyhedron. *Witness: `bt814_tomotope_middle_layer_from_residual_tetrahedra.py`.*

Theorem 5.2 (Photonic mirror middleware). *The full optical Clifford runtime factors as*

$$|\mathrm{Sp}(4, \mathbb{F}_3)| = 51840 = 24 \cdot 2160 = 24 \cdot 45 \cdot 48.$$

Here $24 = f$ is the full $\mathrm{Sp}(4, 3)$ lift of the projective D_{12} mirror-slot stabilizer, 45 is the hyperbolic polar-pair/tritangent geography, and 48 is the local tomatope edge-face middle layer (also the order of the chart group O_h). Equivalently, the photon is clocked by the cyclic selector side and routed by the dihedral mirror side; the complete Clifford group is the lift of that mirror bus through the tomatope middle carrier. Witness: `bt826_photonic_mirror_middleware.py`.

This is the architectural closure missing from a bare universality claim. The tritter, phase plate, and controlled delay generate the whole Clifford group, but the group is *organized* as a middleware stack: full slot stabilizer, global mirror bus, polar vacuum geography, and tomatope local block. In software terms, C_{12} is the clock interrupt, D_{12} is the mirror router, 48 is the local packet format, and $24 \cdot 45 \cdot 48$ is the finite runtime.

6 The software: braids, teleported gates, universality

6.1 Exact braid words

In the anyonic Fibonacci representation $\sigma_1 = \mathrm{diag}(\zeta^4, -\zeta^2)$ ($\zeta = e^{i\pi/5}$) one has *exactly* $\sigma_1^5 = Z$ and $\sigma_1^{10} = I$ — not merely projectively (verified in the cyclotomic ring $\mathbb{Z}[\zeta_{10}]$). In the dual $(\pm)$ encoding of the local $K_{3,3}$ cycle code $\mathbb{F}_2^4$ this makes every register bit-flip an exact five-letter braid word, and the flat global register (the unique connected gluing with trivial holonomy across the atlas) carries the action coherently: a zero-Berry-phase memory. *Witnesses:* `bt740`, `bt741`.

6.2 The photon is its own gate

A gate U is stored as the self-entangled state $(I \otimes U)|\Omega\rangle$; Bell-projecting an input register against the photon's *past* register outputs U (times a known Pauli correction) on the *future* register — verified for all nine outcomes. The machine's gate library is a library of self-entangled states; executing software means interfering the photon with its own history. *Witness:* `bt821`.

6.3 The Universality Theorem

Theorem 6.1 (Clifford completeness from optics). *The symplectic images of the photon's physical elements — the tritter F_3 on either register, the quadratic phase plate S , and the delay-conditioned CX — generate all of $\mathrm{Sp}(4, \mathbb{F}_3)$ (closure of order 51840, exact). Hence tabletop optics realize the complete two-qutrit Clifford group modulo Paulis and phases. Witness: `bt825_universality_theorem.py`.*

Theorem 6.2 (Universality). *Clifford completeness, together with the machine's intrinsic magic supply (§7) and the strict positivity of its contextual fraction (1/10, exact), implies universal quantum computation by magic-state injection; for qutrits, contextuality is necessary and sufficient for magic distillation (Howard–Wallman–Veitch–Emerson). One self-entangled photon on the $W(3, 3)$ mesh is a universal quantum computer whose transport is its gate action.*

7 The fuel: matter = magic

Theorem 7.1 (The triality). *In the photon’s two-qubit reading, exactly the 4 basis rays are stabilizer states and all 36 ω -rays are magic (no stabilizer state has support 3). Consequently three classifications coincide exactly, on rays [4, 36] and on contexts {4:1, 1:12, 0:27}: the holonet vacuum split, the Schmidt (self-entanglement) strata, and the magic strata. The matter shell is the machine’s non-classical resource sector. Witness: bt822.*

Theorem 7.2 (Three grades and the cube inside). *Stabilizer fidelity splits the 36 magic rays into grades*

$$\underbrace{8}_{2^q} \text{ deep } \left(F = \frac{2+\sqrt{3}}{6} \right), \quad \underbrace{24}_f \text{ mid } \left(F = \frac{5+2\sqrt{3}}{12} \right), \quad \underbrace{4}_\mu \text{ shallow } \left(F = \frac{3}{4} = \frac{q}{\mu} \right),$$

the shallow grade sitting exactly at the Werner decoherence threshold. The 8 deep rays are precisely the point set of a hyperbolic polar pair (the 24-cell vacuum axis), and the 27 fully-magic contexts stratify by grade signature as 1+8+12+6 — the cell census of the cube. Witnesses: bt822, bt823.

Theorem 7.3 (Exact contextuality budget). *The maximum number of contexts a classical $\{0, 1\}$ valuation can satisfy exactly-once is exactly $36 = (q!)^2$ (complete branch-and-bound over all 2^{40} markings): classicality saturates at the spread count, the deficit is exactly $\mu = 4$, and the contextual fraction is exactly $1/10 = 1/\Phi_4$. A perfect valuation would be an ovoid, and $W(3, 3)$ has none: the true independence number is $\alpha = 7 = \Phi_6$, attained only by the beacon heptads of §10, against the Lovász bound $\vartheta = 10$ — the $\vartheta - \alpha$ gap $3 = q$ is the combinatorial face of Kochen–Specker. Witnesses: bt818, bt823.*

8 Memory, protection, and the immune system

Theorem 8.1 (Steinberg memory). *The Levi-graph cycle space and the chart-overlap 81-sector are both irreducible and equal to the Steinberg representation of $\mathrm{PSp}(4, 3)$ — the Solomon–Tits homology of the building, whose canonical basis is the 81 apartments through any chamber. Schur’s lemma then forces the uniqueness of the chart–cycle bridge. Restricted to a Sylow 3-subgroup the module is free of rank one: the protected memory is one free generator over the substrate’s ternary symmetry. Witnesses: bt742, bt744.*

The error layer is native: the $[[240, 81, 4, 3]]_3$ CSS code on the edge qutrits has all four parameters substrate-primitive; the dual Császár/Szilassi cross-check supplies parity-free error detection; and the web’s unique non-Ramanujan eigenspace — dimension $15 = g$ — is the spectral sentinel where drift appears first.

9 Time: three clocks and a quasicrystal

Theorem 9.1 (Clock trichotomy). *Of the program’s three natural clocks, exactly one is internal: $\mathbb{Z}_{12}$ (the rectangle clock, $12 \mid 51840$) runs the selector and duo machinery, while $\mathbb{Z}_7$ (Császár) and $\mathbb{Z}_{13}$ (Singer) do not divide the group order and act as external references — both with decimal period $6 = q!$, so the digit expansion of $1/7$ is literally the internal clock’s shadow modulo its duo bit. Witnesses: bt774, bt803, bt807, bt819.*

Theorem 9.2 (The Boerdijk–Coxeter time quasicrystal). *Driving the loop by the BC twist $\theta = \arccos(-2/3)$ per round trip yields a stroboscopic orbit that provably never repeats (Niven), with the Steinhaus three-distance signature at all substrate step counts and — the substrate’s signature moment — exactly two gap lengths at $n = 30 = h(E_8)$, the BC ring length, where closure happens in S^3 (the 600-cell) rather than the phase circle (deficit 0.0158). This is the photonic sibling of the Fibonacci-drive dynamical topological phase. Witness: **bt820**.*

Theorem 9.3 (The arrow of time). *Without its two-trit feedforward record, the photon’s self-configuration loop is the completely depolarizing channel with unique fixed point I/q — exactly the Bell marginal; with the record it is unitary. Decoherence is self-forgetting, priced at $\log_3 q^2 = 2$ trits per cycle. Witness: **bt823**.*

10 Synchronization: beacons and schedules

The maximum classically coherent constellations are the *beacon heptads*: seven states with all pairwise overlaps exactly $1/3 = 1/q$ — a complete K_7 interference mesh, the Császár toroidal node realized in light. There are exactly 2880 heptads in a single transitive family; each has a $\mathbb{Z}_3 \times \mathbb{Z}_3$ symmetry fixing a unique *master beacon* and cycling two triads ($7 = 1 + 3 + 3$), and single-beacon swaps connect the family into its own exchange network: the reference frames of the network form a network. The measurement timetable is rigid: there exist exactly 36 complete schedules (partitions of the 40 states into 10 contexts) — an exhaustive result that simultaneously proves every spread of $W(3,3)$ is regular — with each context appearing in exactly 9 schedules. *Witnesses: **bt817**, **bt819**.*

11 Build sheet and falsifiable witnesses

Components (all standard quantum optics): one heralded single-photon source; one polarizing beam splitter; one symmetric 3-port coupler (tritter); a $0/\tau/2\tau$ delay ladder; one bin-synchronous electro-optic modulator; one polarization rotator set to $\arccos(-2/3)$ in the recirculation loop; single-photon detectors.

Witnesses (kill criteria, all parameter-free):

witness	predicted value	substrate form
trace–Choi visibility of F_3	$1/3$	$1/q$
trace–Choi visibility of X, Z	0	—
Kochen–Specker classical budget	$36/40$	$(q!)^2/v$
beacon-mesh pair visibility	$1/3$ (all 21 pairs)	$1/q$
Werner separability threshold	$3/4$	q/μ
BC-drive gap census at $n=30$	2 gap lengths	$h(E_8)$ ring
D_{12} mirror-slot stabilizer	order 12	projective bus
runtime factorization	$24 \cdot 45 \cdot 48$	$ \mathrm{Sp}(4, \mathbb{F}_3) $

A measured deviation in any row falsifies the corresponding layer.

12 Implications, corrections, and the ethos

Physics. The vacuum is five-fold degenerate, indexed by the Schläfli inventory; the choice $1+12+27$ is a parabolic gauge choice, not a necessity. The generation triples of the matter shell are the

tritangent triangles of the 27-line configuration, reconstructed here purely from double cosets. Matter and magic coincide: the non-classical resource of the machine *is* its matter sector, with the cube’s cell census $1 + 8 + 12 + 6$ grading the fuel.

Computing. Universality requires no exotic hardware: the Clifford layer is three optical elements, the magic layer is the photon’s own matter shell, and the topological layer rides the exact $\sigma^5 = Z$ braid identity. The network-is-computer thesis is not an analogy but the statement that the atlas’s transport groupoid and the register’s gate groupoid are the same groupoid. The middleware theorem adds the missing operating-system layer: the cyclic C_{12} clock orders phase, the dihedral D_{12} mirror bus routes state, the tomatope 48-packet supplies local edge–face incidence, and the full runtime is exactly $24 \cdot 45 \cdot 48 = 51840$.

The ethos. Two pre-existing corpus claims failed under exact computation and were corrected at their sources during this program: the independence number of $W(3, 3)$ is 7, not 10 (no ovoid exists, the “perfect graph” block is withdrawn), and the Kochen–Specker optimum is $36/40$, not $34/40$. In both cases the corrected values are *more* substrate-natural than the claims they replaced ($\alpha = \Phi_6$; budget = $(q!)^2/v$) — the pattern a real structure shows when probed honestly.

A Verification ledger

Every theorem above is backed by an executable script in **analysis/** with results in **data/**; the chain BT739–BT826 was developed and pushed incrementally with exact arithmetic ($\text{GF}(p)$ elimination, cyclotomic integers, complete branch-and-bound, GAP witnesses for group-theoretic facts).

layer	primary witnesses
duality / carriers	bt817, bt820, bt821
atlas / building / web	bt773, bt777, bt744, bt779
mirror middleware / tomatope runtime	bt814, bt815, bt826
braids / registers	bt740, bt741
vacua / geography / transitions	bt808–bt813, bt816
fuel / contextuality	bt818, bt822, bt823
memory / Steinberg	bt742, bt744
clocks / quasicrystal / arrow	bt774, bt819, bt820, bt823
sync / schedules	bt817, bt819
universality	bt825
corpus corrections	bt818, bt823, bt824

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